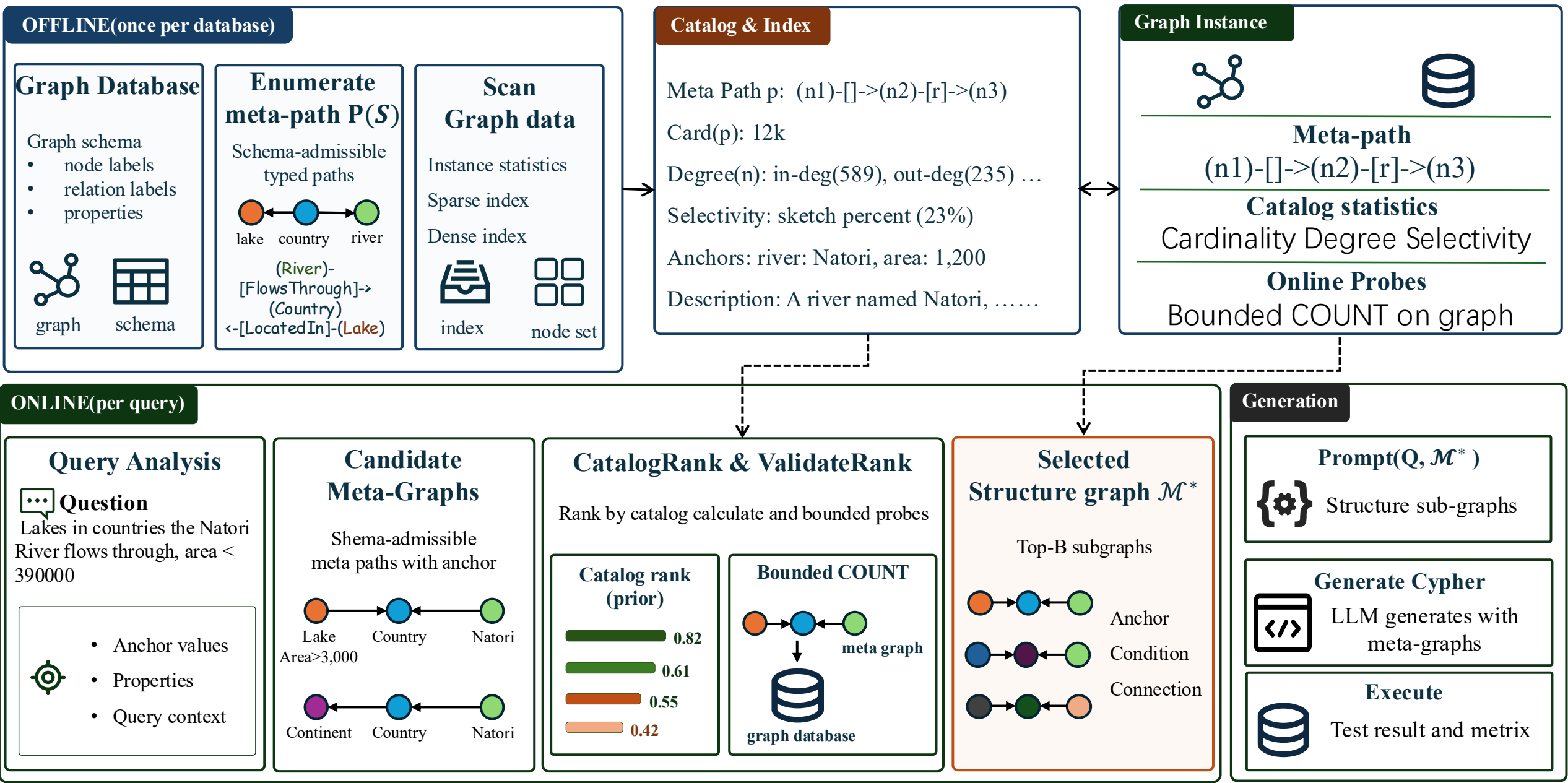

OFFLINE(once per database)
Graph Database
Graph schema
• node labels
• relation labels
• properties
graph
schema
Enumerate meta-path P(S)
Schema-admissible typed paths
lake
country
river
(River)-[FlowsThrough]->(Country)<-[LocatedIn]-(Lake)
Scan Graph data
Instance statistics
Sparse index
Dense index
index
node set
Catalog & Index
Meta Path p: (n1)-[]->(n2)-[r]->(n3)
Card(p): 12k
Degree(n): in-deg(589), out-deg(235) …
Selectivity: sketch percent (23%)
Anchors: river: Natori, area: 1,200
Description: A river named Natori, ……
Graph Instance
Meta-path
(n1)-[]->(n2)-[r]->(n3)
Catalog statistics
Cardinality Degree Selectivity
Online Probes
Bounded COUNT on graph
ONLINE(per query)
Query Analysis
Question
Lakes in countries the Natori River flows through, area < 390000
• Anchor values
• Properties
• Query context
Candidate Meta-Graphs
Shema-admissible meta paths with anchor
Lake
Area>3,000
Country
Natori
Continent
Country
Natori
CatalogRank & ValidateRank
Rank by catalog calculate and bounded probes
Catalog rank (prior)
0.82
0.61
0.55
0.42
Bounded COUNT
meta graph
graph database
Selected Structure graph $\mathcal{M}^*$
Top-B subgraphs
Anchor
Condition
Connection
Generation
Prompt(Q, $\mathcal{M}^*$)
Structure sub-graphs
Generate Cypher
LLM generates with meta-graphs
Execute
Test result and metrix